# CST583 - Advanced Data Science

## CST 583 (Advanced Data Science) Syllabus

### Course Information

#### Course Details

- **Course Title:** Advanced Data Science
- **Course Number:** CST 583
- **Meeting Location:** Physical
- **Units:** 4

#### Course Description

This course covers key topics in data science, including data preparation, exploratory data analysis, statistical techniques, machine learning, and visualization. Students will learn how to manage, analyze, and interpret large datasets using modern data science tools.

#### Materials

- Course textbook: "Data Science from Scratch" by Joel Grus

#### Technology Requirements

Students must have access to: - A computer with reliable internet connection - Access to Canvas learning management system

#### Course Objectives

At the end of this class, you should be able to:

1. Perform data wrangling and preprocessing on large datasets
2. Apply statistical and machine learning models to real-world problems
3. Use visualization tools to communicate insights
4. Critically evaluate data science methods and their limitations

### Grading

| Assignment Kind | Value |
|---|---|
| Projects | 40% |
| Exams | 30% |
| Participation | 30% |

### Projects

Projects involve data analysis, application of machine learning models, and the creation of data visualizations. Students will complete hands-on assignments using real datasets.

### Exams

Exams will test both conceptual understanding and practical skills in data science, covering data wrangling, machine learning, and statistical methods.

### Participation

Participation includes involvement in discussions, peer collaboration, and sharing insights during in-class exercises. Attendance is required.

### Grade Assignment

| Grade Range | Letter Grade |
|---|---|
| $[97, \infty)$ | A+ |
| $[93, 97)$ | A |
| $[90, 93)$ | A- |
| $[87, 90)$ | B+ |
| $[83, 87)$ | B |
| $[80, 83)$ | B- |
| $[77, 80)$ | C+ |
| $[73, 77)$ | C |
| $[70, 73)$ | C- |
| $[60, 70)$ | D |
| $[0, 60)$ | F |

## Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion. Students will be notified of any substantive changes via Canvas announcement and/or email.